

LYMPHOID NEOPLASIA

Deregulation of *FOXF1/FENDRR* from *t*(14;16)(q32;q24) defines a subtype of high-risk lineage ambiguous leukemia

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KEY POINTS

- *t*(14;16) defines a subtype of high-risk lineage ambiguous acute leukemia with *FOXF1/FENDRR* activation and *GATA3* mutation.
- *t*(14;16) leukemia exhibits sensitivity to BCL-2 and JAK/STAT inhibition.

Despite great progress in understanding the genomic basis of immature T-cell acute lymphoblastic leukemia/lymphoblastic lymphoma (T-ALL) and acute leukemias of ambiguous lineage, there are still cases that lack defining genetic markers, complicating risk stratification, and limiting targeted therapeutic options. Recent studies have shown that enhancer hijacking drives oncogene activation in approximately half of T-ALL cases, with the *BCL11B* (*BCL11* transcription factor B) enhancer frequently involved. Here, we describe a subtype of leukemia with a distinct gene-expression signature, and immunophenotype characterized by positivity for immature (CD38), myeloid (CD13), T-lymphoid (cytoplasmic [c]CD3, CD7), and B-lymphoid markers (CD19, CD79a, CD10). This subtype is defined by the *t*(14;16)(q32;q24) translocation, which places the *FOXF1* gene and its antisense long noncoding RNA gene *FENDRR* under the regulatory control of the *BCL11B* enhancer, leading to their ectopic transcriptional activation. Common concomitant

genetic lesions are loss-of-function alterations of *GATA3*, *CDKN2A/CDKN2B* deletion and activating JAK/STAT and NOTCH1 pathway mutations. Patients were predominantly children and adolescents/young adults and experienced poor treatment outcome. High-throughput drug screening of 176 compounds demonstrated efficacy of combined BCL2 apoptosis regulator family proteins and JAK/STAT signaling inhibitors. Additionally, the clinical use of tyrosine kinase inhibitors in some of these patients showed therapeutic efficacy. Collectively, these findings identify *BCL11B*-enhancer-mediated deregulation of *FOXF1/FENDRR* as a hallmark of a subtype of high-risk lineage ambiguous leukemia that is potentially amenable to targeted therapeutic intervention.

Introduction

Acute leukemia has traditionally been classified using immunophenotypic and genetic features. However, this approach remains limited when applied to acute leukemias of ambiguous lineage, either due to incomplete lineage commitment or

coexpression of markers from multiple hematopoietic lineages, such as mixed-phenotype acute leukemia (MPAL) and early T-cell precursor acute lymphoblastic leukemia (ETP-ALL). MPAL is characterized by coexpression of markers of multiple hematopoietic lineages most commonly myeloid with T- and/or B-cell markers.¹ ETP-ALL is characterized by positivity for

cytoplasmic CD3, negativity for other T-cell antigens (CD1a, CD8, and weak/absent CD5) and the aberrant expression of myeloid or stem cell markers.^{2,3} Recent genomic studies have identified multiple new drivers and oncogenic mechanisms, new subtypes of lineage ambiguous leukemia, and expanded the definition of ETP-ALL to “ETP-like” ALL that includes additional T-ALL cases that do not fulfill the immunophenotypic criteria for ETP-ALL but have similar gene-expression profile and driver alterations.⁴

Additionally, several genomic alterations have been identified that transcend immunophenotypic categories, such as *ZNF384* rearrangements in B-ALL and B/myeloid MPAL,^{5,6} and *BCL11B* (*BCL11* transcription factor B) activated leukemia, in which rearrangements of hematopoietic stem cell enhancers to *BCL11B* activate its expression in a subset of T/myeloid MPAL, ETP-ALL and AML.^{7,8} Conversely, the *BCL11B* enhancer is one of the most hijacked regulatory elements in T-ALL,⁴ being rearranged to the *TLX3*, *HOXA*, *NKX2-1*, and *NKX2-5* oncogenes,⁹⁻¹² as well as *LMO2*, *HOXB13*, *IL7R*, and *NOTCH1*.⁴

Here we report a new subtype of immature T-ALL and acute leukemias of ambiguous lineage, characterized by a t(14;16)(q32;q24) translocation resulting in the transcriptional activation of *FOXF1* and *FENDRR*, through hijacking of the *BCL11B* enhancer.

Materials and methods

Patient cohorts and genomic profiling

We retrospectively investigated patients with T-ALL and MPAL from 2 independent cohorts of 1135 non-ETP ALL, 398 ETP/near-ETP ALL, 19 MPAL, and 279 unknown cases (337 adults and 1494 children) (supplemental Table 1, available on the *Blood* website): 510 samples from the University of Perugia (cohort 1) screened by cytogenetics and 1321 cases from St. Jude Children's Research Hospital (cohort 2) analyzed by whole-genome sequencing (WGS). Genomic profiling of samples harboring the t(14;16)(q32;q24) was performed using fluorescence in situ hybridization (FISH; supplemental Table 2), WGS, single nucleotide polymorphism (SNP) array, and custom-targeted next generation sequencing.¹³ Hi-C and chromatin immunoprecipitation (HiChIP) and assay for transposase-accessible chromatin using sequencing (ATAC-seq) were performed in samples from patients SJALL079622 and SJTALL005179 (Table 1).

Expression studies and PDX generation

RNA sequencing (RNA-seq) was performed on 10 t(14;16) samples and integrated with published T-ALL data sets⁴ for dimensionality reduction, differential expression, gene set enrichment, and allele-specific expression analyses. Patient-derived xenograft (PDX) models were established from 3 samples (SJALL080600, SJALL079622, and SJTALL005179 in Table 1), and western blotting and ChIP-seq experiments were carried out on 2 to evaluate *FOXF1* expression and chromatin occupancy. A cell line was established from the SJALL079622-derived PDX and subjected to CRISPR/Cas9-mediated *FOXF1* inactivation to assess its functional relevance.

Single-cell analysis of differentiation and clonal composition

Differentiation stage and clonal architecture were investigated by single-cell RNA sequencing (scRNA-seq) on diagnostic bone

marrow (BM) from SJALL079622 and SJTALL005179 and their corresponding PDXs, as well as on mice engrafted with purified SJALL079622-derived populations. Targeted single-cell DNA and protein sequencing was performed on the primary and PDX samples from SJALL079622.

Drug-resistance profiling and functional validation assays

Ex vivo drug response profiling was performed on 2 PDX models using a 176-compound library. Promising molecules were further validated through dose-response and synergy assays. BH3 mimetic profiling and phosphoflow experiments were used to corroborate drug-induced apoptotic dependencies and signaling responses.

Full details of materials and methods are provided in the supplemental Materials and Methods.

Results

Identification of a novel t(14;16)(q32;q24) in T-ALL and MPAL

A previously uncharacterized t(14;16)(q32;q24) was detected in samples from 13 patients with acute leukemia (Table 1; supplemental Table 3). Among the 73 cohort 1 patients with 14q32.3/*BCL11B* alterations, 7 harbored 16q24 rearrangements (Figure 1A-B; supplemental Table 4). WGS on a second cohort revealed 6 samples with rearrangement between 14q32 and 16q24 comparable to the molecular cytogenetic results from cohort 1 (supplemental Tables 5-7).

The 14q32 breakpoints were located 70-600 kb telomeric to *ThymoD*, a regulatory element required for *BCL11B* activation during thymocyte development,^{14,15} resulting in the relocation of *BCL11B* to a derivative chromosome 16 (Figure 1C). The 16q24 breakpoints were 26 to 356 kb centromeric to the transcription factor gene *FOXF1*, and in 1 case (SJALL080600), within its adjacent lncRNA *FENDRR* (Figure 1D). Consequently, *FOXF1* and *FENDRR* were relocated to the derivative chromosome 14, telomeric to the *BCL11B* enhancer.

The t(14;16)(q32;q24) was cryptic in 12 of 13 cases (Table 1) and was mutually exclusive with all established cytogenetic subtype-defining alterations in T-ALL, suggesting it is the primary oncogenic driver alteration. Concomitant copy number abnormalities involving both der(14) and der(16) were identified in 4 samples (supplemental Tables 4 and 8; Figure 1E-F).

Chromatin reorganization of the *BCL11B* and *FOXF1/FENDRR* loci

Combined histone 3 lysine 27 acetylation (H3K27ac) HiChIP on 2 samples (SJALL079622 and SJTALL005179) identified interaction between multiple *BCL11B* regulatory elements and the *FOXF1/FENDRR* locus on der(14) (supplemental Figure 1A-B), consistent with an enhancer hijacking mechanism. These elements include *ThymoD* and the long noncoding *LINC01550*, which formed loops with the *FOXF1/FENDRR* locus in both samples (Figure 1G-H). In contrast, loops between *BETA*, a noncoding region amplified in a subset of *BCL11B*-leukemia,⁸ and *FOXF1/FENDRR* locus were observed only in SJALL079622, where *BETA* remains on der(14) (Figure 1G). Only in SJALL005179, which

Table 1. Clinical characteristics of patients with *FOXF1/FENDRR* ALL

Sample	Sex/ age, y	Initial Dx	WBC ($\times 10^9$ /L)	Immunophenotype/TCR status	Karyotype	Extramedullary localization	Treatment (response, months from diagnosis)
SJALL080602 (UPN717)	M/26	ETP-ALL	23.5	CD38⁺, CD13⁺, cyCD3⁺, CD10⁺, cyCD79a⁺, CD7⁺, CD5⁺, CD99⁺ CD34⁺, CD4⁺, CD1a⁺ , TdT ⁺ , CD8 ⁺ , CD3 ⁺ , CD2 ⁺ , CD11b ⁺ , CD19 ⁺ , CD33 ⁺ , CD56 ⁺ , CD65 ⁺ , CD66b ⁺ , CD117 ⁺ , HLA-DR ⁺ , CD22 ⁺ , CD79a ⁺ , MPO ⁺ TCR: not rearranged	47,XY,+3,add(4p15),add(12p13)[2] 47,idem,add(9q34), ?del(14q32)[3] 46,XY[17]	CNS LN (diffuse) nasopharynx, bone (ilium, ribs, vertebrae)	GIMEMA LAL1913 C1- C7 ⁴⁶ (partial remission, relapse +10) daratumumab (refractory) Dasatinib (remission +2.5, relapse, died +16)
SJALL080601 (UPN716)	F/24	ETP-ALL	13	CD38⁺, CD13⁺, cyCD3⁺, CD10⁺, cyCD79a⁺, CD7⁺, CD5⁺ CD34⁺, CD4⁺, CD1a⁺ , TdT ⁺ , CD8 ⁺ , CD3 ⁺ , CD2 ⁺ , CD117 ⁺ , CD33 ⁺ , HLA-DR ⁺ , MPO ⁺ , CD19 ⁺ , CD14 ⁺ TCR: N.A.	N.A.	mediastinum LN (axillary)	PETHEMA ALL96 ⁴⁷ (refractory) HAM I (partial remission) HAM II (partial remission, progression, died +7)
SJALL080600 (UPN1281)	M/37	ETP-ALL	18.2	CD13⁺, cyCD3⁺, cyCD79a⁺, CD7⁺, CD5dim, CD2⁺, TdT⁺ CD34⁺, CD1a⁺, CD4⁺ , CD117 ⁺ , CD8 ⁺ , CD3 ⁺ , HLA-DR ⁺ , CD10 ⁺ , CD19 ⁺ , CD33 ⁺ , CD15 ⁺ , CD14 ⁺ , CD56 ⁺ TCR: not rearranged	N.A.	mediastinum (bulky) LN (diffuse) tonsil	GIMEMA LAL1913 C1 ⁴⁶ (partial remission) GIMEMA LAL1913 C2- C8 ⁴⁶ (remission, relapse +23) GIMEMA LAL1913,C1 ⁴⁶ (refractory) FLAI (remission) HSCT (remission, relapse +26, died +38)
SJALL080606 (UPN809)	M/18	B/T MPAL	70.5	CD38⁺, CD13⁺, cyCD3⁺, CD19dim, CD10⁺, cyCD79a⁺, CD7⁺, CD2⁺, CD5⁺dim CD34⁺, CD4⁺ , CD117 ⁺ , CD1a⁺ , TdT ⁺ , CD8 ⁺ , CD3 ⁺ , CD33 ⁺ , MPO ⁺ , CD56 ⁺ , CD20 ⁺ , CD22 ⁺ , CD61 ⁺ TCR: not rearranged	N.A.	CNS mediastinum (bulky) pleural effusion	GIMEMA LAL1913 C1 ⁴⁶ (refractory) MTX+ARA-C (refractory) mitoxantrone(refractory) Vinblastine+etoposide (refractory) FLAI (refractory) nelarabine (refractory) Haplo-HSCT (died +21)
SJALL080605 (UPN721)	M/29	ETP-ALL	11.5	CD38⁺, CD13⁺, cyCD3⁺, CD19⁺, CD10⁺, cyCD79a⁺, CD7⁺, CD8⁺, CD117⁺, CD45⁺, TCRA⁺ CD34⁺, CD4⁺ , TdT ⁺ , CD3 ⁺ , CD2 ⁺ , CD5 ⁺ , CD33 ⁺ , MPO ⁺ , HLA-DR ⁺ , CD20 ⁺ , cCD22 ⁺ , CD14 ⁺ , CD64 ⁺ , CD123 ⁺ , CD66b ⁺ , CD11b ⁺ , TCRG ⁺ , CD138 ⁺ TCR: not rearranged	N.A.	mediastinum (bulky) spleen pleural effusion	GIMEMA LAL1913 C1 ⁴⁶ (refractory) GIMEMA LAL1913 C2 ⁴⁶ (partial remission) GIMEMA LAL1913 C3 ⁴⁶ (remission) HSCT (remission, Alive +48)

Shared immunophenotypic markers are in bold.

ARA-C, cytarabine; CNS, central nervous system; CTX, cyclophosphamide; Initial Dx, initial diagnosis provided by the referring Institution; DLI, donor lymphocyte infusion; FLAI, fludarabine, cytarabine, idarubicin; HAM, High-dose cytosine arabinoside and mitoxantrone; Haplo, haploidentical; HSCT, hematopoietic stem cell transplantation; ICE, idarubicin, cytarabine and etoposide; IDA, idarubicin; LN, lymph nodes; MTX, methotrexate; MITO, mitoxantrone; MUD, matched unrelated donor; N.A., not available; PEG-ASP, peg-asparaginase; Remission, hematological and/or cytogenetic remission.

Table 1 (continued)

Sample	Sex/ age, y	Initial Dx	WBC ($\times 10^9/L$)	Immunophenotype/TCR status	Karyotype	Extramedullary localization	Treatment (response, months from diagnosis)
SJALL080603 (UPN715)	M/28	ETP-ALL	184.2	CD38⁺, CD13⁺, cyCD3⁺, CD19⁺, CD10⁺ , CD7⁺ , CD2 ⁺ , CD5 ⁺ partial, CD117 ⁺ partial, CD34⁺, CD1a⁺, CD4⁺ , CD8 ⁺ , TdT ⁺ , CD3 ⁺ , HLA-DR ⁺ , cCD79a ⁺ , CD20 ⁺ , CD22 ⁺ , CD25 ⁺ TCR: N.A.	46,X,dic(Y;16)(q11.2;p13),t(1;19) (p36;p13),inv(5)(p13q31), t(14;16)(q32;q24),+der(16)t(14;16) (q32;q24),r(18)(p11.2q23)[22]	CNS mediastinum (bulky) LN (diffuse) tonsil	IDA+PEG-ASP; MTX+ARA-C (remission +12) MUD-HSCT (relapse) nelarabine+DLI (progression, died +17)
SJALL080604 (UPN714)	M/22	T/myeloid MPAL	31.3	Population 1 (45%): CD38⁺, CD13⁺, cyCD3⁺, CD7⁺ , CD2 ⁺ , TdT ⁺ , CD34⁺, CD4⁺ , CD5 ⁺ , CD10 ⁺ , CD11b ⁺ , CD14 ⁺ , CD33 ⁺ , CD117 ⁺ , HLA- DR ⁺ , CD56 ⁺ , MPO ⁺ , CD14 ⁺ , CD64 ⁺ , cyCD79a ⁺ , CD19 ⁺ , CD20 ⁺ , Population 2 (13%): CD38⁺, CD13⁺ , MPO ⁺ , CD33 ⁺ , CD64 ⁺ , CD117 ⁺ ; CD34⁺, CD4⁺ , TdT ⁺ , cyCD3 ⁺ , CD2 ⁺ , CD5 ⁺ , CD7 ⁺ , CD10 ⁺ , CD11b ⁺ , CD14 ⁺ , CD20 ⁺ , HLA-DR ⁺ TCR: rearranged	46,XY,t(1;13)(q12;p13), t(16;21)(p13;q22),del(17q)[24] 46,XY[1]	Mediastinum (bulky) LN (axillary) pleural effusion	ICE-FLAG-MITO-HCVAD (refractory) GIMEMA LAL1913 ⁴⁶ (refractory) nelarabine- CTX (refractory) Ponatinib (partial remission +3 weeks) HSCT (remission +4, died +11)
SJALL079622	M/11	B/T MPAL	165	CD38⁺, CD13⁺, cyCD3⁺, CD19⁺, cyCD79a⁺ , CD3 ⁺ , CD2 ⁺ , CD8 ⁺ CD34⁺, CD1a⁺, CD4⁺ , TdT ⁺ , CD5 ⁺ , CD22 ⁺ , CD33 ⁺ , MPO ⁺ , CD10 ⁺ , HLA-DR ⁺ , CD117 ⁺ TCR: not rearranged	46,XY,der(5)t(5;13)(q32;q32), der(13)t(5;13)(q31;q14)	N.A.	JPLSG T-11 ⁴⁸ (refractory) AML-directed therapy (refractory) HSCT (died +7)
SJALL079532	M/11	B/T MPAL	9.6	cyCD3⁺, CD13⁺, CD19⁺, cyCD79a⁺ , CD2 ⁺ , CD33 ⁺ , CD117 ⁺ , cy-MPO ⁺ , CD5 ⁺ , CD1a⁺, CD34⁺, CD4⁺ , TdT ⁺ , CD8 ⁺ , sCD3 ⁺ , CD10 ⁺ , CD22 ⁺ , HLA-DR ⁺ TCR: not rearranged	46,XY,add(6)(p21),add(9)(p13),-11,+mar[19] 46,XY[1]	CNS LN (diffuse)	NHL-BMF-95 (refractory) AML-directed therapy (refractory) HSCT (died +12)
SJTALL003	M/7	ETP-ALL	44.8	cyCD3⁺, CD13⁺, cyCD79⁺ , CD2 ⁺ , CD8 ⁺ CD34⁺, CD1a⁺, CD4⁺ , TdT ⁺ , CD10 ⁺ , CD19 ⁺ , CD117 ⁺ , CD5 ⁺ , CD33 ⁺ , MPO ⁺ , HLA-DR ⁺ , CD3 ⁺ TCR: N.A.	46,XY,del(1)(p22p32),inv(2)(p11.2q13)c, der(4)t(4;10)(q21;q26),der(5)t(5;10) (p13;p11.2), der(10)t(4;10;5)(q?;p11.2;p13)[18] 46,XY,inv(2)(p11.2q13)c[2]	CNS	Total XV ⁴⁹ (refractory, died +11.7)
SJTALL007	M/18	ETP-ALL	13.1	CD13⁺, cyCD3⁺, CD10⁺, cyCD79⁺ , CD2 ⁺ , CD3 ⁺ , CD5 ⁺ , TdT ⁺ , CD1a⁺, CD34⁺, CD4⁺ , CD8 ⁺ , CD19 ⁺ , CD33 ⁺ , CD117 ⁺ , MPO ⁺ , HLA-DR ⁺ TCR: not rearranged	46,XY,+6,inv(10)(p13q22), add(12)(p13),der(12)t(12;14)(p13;q11.2), del(13)(q22q24),-14[17]/46,XY [3]	CNS	Total XV ⁴⁹ (refractory, died +44.8)

Shared immunophenotypic markers are in bold.

ARA-C, cytarabine; CNS, central nervous system; CTX, cyclophosphamide; Initial Dx, initial diagnosis provided by the referring Institution; DLI, donor lymphocyte infusion; FLAI, fludarabine, cytarabine, idarubicin; HAM, High-dose cytosine arabinoside and mitoxantrone; Haplo, haploidentical; HSCT, hematopoietic stem cell transplantation; ICE, idarubicin, cytarabine and etoposide; IDA, idarubicin; LN, lymph nodes; MTX, methotrexate; MITO, mitoxantrone; MUD, matched unrelated donor; N.A., not available; PEG-ASP, peg-asparaginase; Remission, hematological and/or cytogenetic remission.

Table 1 (continued)

Sample	Sex/ age, y	Initial Dx	WBC ($\times 10^9/L$)	Immunophenotype/TCR status	Karyotype	Extramedullary localization	Treatment (response, months from diagnosis)
SJTALL005179	F/19	ETP-like ⁴	73	N.A. TCR: not rearranged		N.A.	COG AALL03B1 ⁵⁰ (refractory) COG AALL0434 arm D ⁵¹ (refractory) HSCT (died +10)
SJALL070948	F/17	ETP-like ⁴	74	N.A. TCR: not rearranged		N.A.	COG AALL03B1 ⁵⁰ (refractory) COG AALL0434 arm D ⁵¹ (refractory, died +5)

Shared immunophenotypic markers are in bold.
ARA-C, cytarabine; CNS, central nervous system; CTX, cyclophosphamide; Initial Dx, initial diagnosis provided by the referring institution; DLI, donor lymphocyte infusion; FLA, fludarabine, cytarabine and etoposide; IDA, idarubicin, cytarabine and etoposide; ICE, idarubicin, cytarabine and etoposide; MUD, matched unrelated donor; N.A., not available; PEG-ASP, peg-asparaginase; Remission, hematological and/or cytogenetic remission.

harbors a more centromeric breakpoint, BETA was relocated to der(16) where it continued to interact with the *BCL11B* promoter (supplemental Figure 1C-D). Interestingly, although all these regulatory elements are active in the t(14;16) samples, they showed distinct activation kinetics in normal T-cell differentiation stages.¹⁶ BETA and ThymoD are active from the early thymic precursor (CD34⁺CD1a⁻) to the immature single positive thymocyte (CD34⁺, CD1a⁺, and CD4⁺). In contrast, *LINC01550* and ThymoD sustain *BCL11B* expression in later stages of hematopoietic differentiation (supplemental Figure 1E). In these 2 samples they were all active suggesting arrest at an early developmental stage.

The *FOXF1/FENDRR* locus was marked by H3K27 acetylation in t(14;16) samples but showed no activity in cord blood (cb) CD34⁺ cells or during thymic differentiation (supplemental Figure 1F). However, ATAC-seq revealed an open chromatin profile at the *FOXF1/FENDRR* locus in the t(14;16) subtype, as well as in cb-derived precursors and the earliest stages of thymic T-cell differentiation, with a marked reduction in accessibility starting from the double-positive precursor stage onward (supplemental Figure 1G). This accessibility pattern suggests the possibility of a translocation event at an early stage of development. Consistent with its tissue specificity, the lung-specific enhancer of *FOXF1* remains inactive in normal cbCD34⁺ cells, thymic differentiating precursors, and t(14;16) samples (supplemental Figure 1F).

t(14;16)(q32;q24) leukemia exhibits a distinct gene-expression profile

Analysis of RNA-seq data showed clustering of t(14;16) samples distinct from other subtypes of T-ALL (Figure 2A; supplemental Table 9), upregulation and allele-specific expression of *FOXF1* and *FENDRR*, and downregulation of *BCL11B* (Figure 2B-D; supplemental Table 10). The breakpoint in *FENDRR* was intragenic in case SJALL080600, and accordingly, this case expressed only the short isoform of *FENDRR* (NR_036444.1) (supplemental Figure 2A). Due to the hallmark expression of these genes, we hereafter refer to this subtype of acute leukemia as “*FOXF1/FENDRR*” ALL. Accordingly, other oncogenic drivers of ETP/ETP-like were not specifically associated with this subtype (supplemental Figure 2B). Five ETP-like samples showed high expression of *FOXF1* and/or *FENDRR* without harboring the t(14;16) (supplemental Figure 2C), with allele-specific expression of *FOXF1* detected in one harboring a 16q loss (supplemental Figure 2D), suggesting that alternative mechanisms may drive the activation of these genes in the absence of the fusion event.

Activation of epithelial-mesenchymal transition, inflammatory, and immune-related pathways, JAK/STAT signaling (Figure 2E; supplemental Table 11), as well as enrichment of myeloid transcriptional programs (supplemental Table 12), was observed in association with the t(14;16) leukemic signature. An intrinsic monocytic program, rather than low tumor purity, likely accounts for the proximity of *FOXF1/FENDRR* ALL samples to the TME-enriched subgroup⁴ (supplemental Figure 2E-F). Indeed, exclusion of TME-enriched samples results in the preferential clustering of *FOXF1* samples in closer proximity to the ETP-like subtype, highlighting their shared progenitor-like features (supplemental Figure 2G).

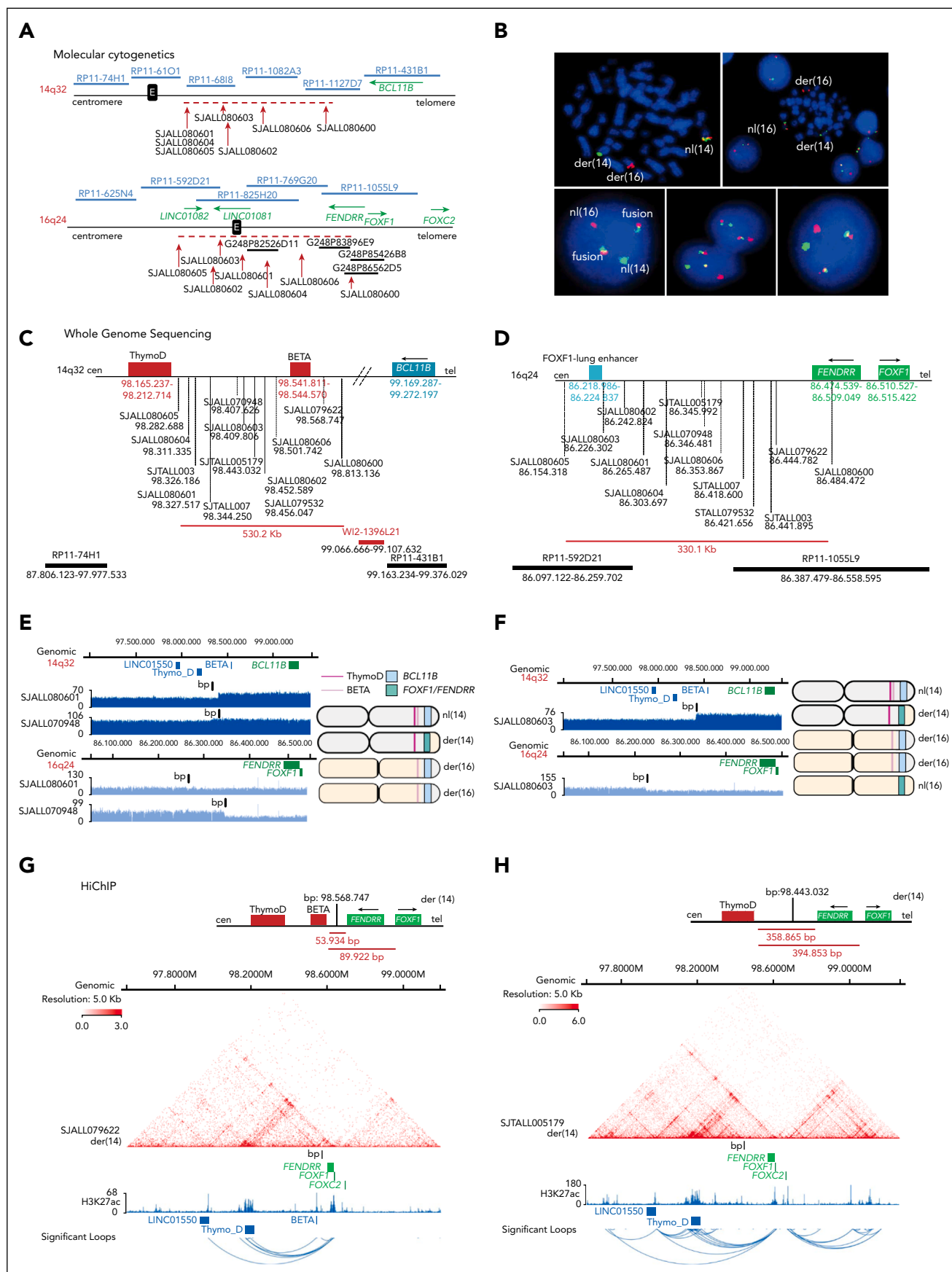


Figure 1.

Genomic features in *FOXF1/FENDRR* ALL

GATA3 mutations affecting the DNA-binding domain were identified in all 13 samples (Figure 2F-G). Four (SJALL080600, SJALL080606, SJALL080605, and SJALL0806041) harbored multiple mutations, and in 2 of these (SJALL080605 and SJALL080604), they were located in close proximity but on different alleles, indicating a compound heterozygous state (supplemental Tables 13-18). Three additional samples (SJALL080601, SJALL080603, and SJALL003), had a concomitant GATA3 deletion (Figure 2F). Collectively, biallelic inactivation of GATA3 was demonstrated in 5 of 13 samples. Although GATA3 mutation is common to all samples, it is not the hallmark of this subtype as it can be detected in non-*FOXF1/FENDRR* T-ALL (Figure 2H). However, given its physiological role in promoting T-cell differentiation and restricting non-T-lineage development,¹⁷ we analyzed the expression of key GATA3 target genes involved in T-lymphoid commitment in leukemic cells. Consistently, we observed downregulation of GATA3 targets driving the T-lineage specification,¹⁸ such as *IKZF2*, *DNTT/TdT*, and *BCL11B* (Figure 2C,I). By contrast, we found upregulation of genes that are downregulated following GATA3 overexpression in thymic CD34⁺ CD1a⁻ progenitors¹⁸ (supplemental Figure 2H), including *SPI1/PU.1* and *ID2* (Figure 2I), indicating that GATA3 alterations exert a key leukemogenic role in this subtype of leukemia.

FOXF1/FENDRR ALL samples with t(14;16) also harbored genomic alterations associated with typical T-ALL. All samples harbored biallelic *CDKN2A/B* deletions and mutations in JAK/STAT signaling genes (Figure 2F; supplemental Tables 8 and 16), consistent with pathway activation observed by gene set enrichment analysis (GSEA; Figure 2E; supplemental Table 11). *NOTCH1/FBXW7* mutations and/or deletions were present in 12 samples, whereas *ETV6* deletions and *RUNX1* rearrangements/mutations were found in 7 and 3 samples, respectively (Figure 2F; supplemental Tables 8 and 16).

Clonal and transcriptional heterogeneity in *FOXF1/FENDRR* leukemia reflects early progenitor arrest and lineage plasticity

Projection of the *FOXF1/FENDRR* gene-expression signature from bulk RNA-seq onto the scRNA-seq data from normal thymus and BM samples¹⁹ revealed enrichment of BM and early thymic precursor signatures, also suggesting developmental arrest at an early progenitor stage. Downregulation of T-cell signatures, combined with the prominence of myeloid,

natural killer (NK), and mature B-cell gene signatures, supports a shift toward non-T-lineage programs (Figure 3A).

Notably, projection of scRNA-seq data of 2 samples onto the normal reference map¹⁹ revealed marked both intratumoral and intersample transcriptomic variability (Figure 3B). Sample SJTALL005179 displayed a homogeneous pro-T transcriptional signature, whereas SJALL079622 was enriched for cells with a transcriptional profile resembling early hematopoietic progenitors, indicative of an immature and developmentally arrested state, alongside a more differentiated monocyte-like population. Clustering analysis identified 2 major distinct clones enriched for myeloid- (My clone) or B and T lymphoid (B/T clone)-associated genes (Figure 3C; supplemental Figure 3A), corroborating immunophenotypic and morphological evidence of dual-lineage blasts (Figure 3D-E). Notably, key genes from the t(14;16)-associated signature, such as *FOXF1*, *FENDRR*, *CD79A*, and *CD19*, were enriched in the lymphoid subset (Figure 3F), whereas both clones harbored the t(14;16) translocation as confirmed by WGS (supplemental Tables 19-22) and FISH (data not shown) on sorted-cell populations, demonstrating an early origin and the role of cell context in determining the gene-expression signature (supplemental Tables 20-22). Only the lymphoid clone, expressing *FOXF1/FENDRR*, sustained engraftment in immunocompromised mice (Figure 3G-H). Interestingly these cells were able to repopulate the myeloid compartment as previously showed for other cases of MPAL,⁵ further demonstrating the intrinsic lineage plasticity of those cells (Figure 3I). Consistent with this observation, GSEA revealed that the B/T lymphoid clone exhibits enrichment of stem-like and immature programs, together with reduced expression of mature myeloid and monocytic signatures (supplemental Figure 3B). Accordingly, scRNA-seq from mice injected with the B/T clone detected only a small minority of mature myeloid cells which showed low *FOXF1* and *FENDRR* expression, supporting the context-dependent regulation of these genes and limited expansion potential (supplemental Figure 3C). Clonal selection of pro-T cells and a marked loss of immature and monocytic populations were observed in SJTALL079622-derived PDX (supplemental Figure 3D). Single-cell DNA sequencing showed biallelic loss of 9p21.3/*CDKN2A* to be followed by a GATA3 mutation and subsequent *IL7R* mutation (supplemental Figure 3E). Collectively, these data suggest a model in which *FOXF1/FENDRR* deregulation is the primary oncogenic event, with GATA3 inactivation restricting T-lineage maturation and enabling lineage plasticity, and in which lymphoid differentiation is associated with higher *FOXF1* expression and propagation of the leukemic clone.

Figure 1. Structural genomic architecture of the t(14;16) rearrangement. (A) Break-apart CI-FISH probe panels used for refining the 14q32 (top panel) and 16q24 (bottom panel) chromosomal breakpoints. Blue and black horizontal lines denote genomic clones and fosmids (GRCh37/hg19), respectively. Genes are depicted in green, with their orientation indicated by the direction of the arrow. Red markers indicate the position of the breakpoint for each case, while the dotted red line represents the breakpoint region on each chromosome. Both panels are not to scale. (B) FISH results from sample SJALL080602 (UPN529). Top: the left panel shows the *BCL11B*/14q32 break-apart assay (RP11-74H1 green/RP11-431B1 orange) on metaphases, indicating 1 fusion (nl(14)), 1 green (der(14)), and 1 orange (der(16)) hybridization pattern. In the right panel the *FOXF1*/16q24 break-apart assay (RP11-625N4 orange/RP11-1055L9 green) on metaphase and nuclei showing relocation of RP11-1055L9 on der(14) (green signal). Bottom: nuclei hybridized with double color-double fusion assay RP11-74H1/RP11-431B1 green + RP11-625N4/RP11-1055L9 orange. The double fusion signal identifies the t(14;16). (C) Genomic breakpoints on chromosome 14q32 were identified in a ~530.2 kb region between the ThymoD enhancer and *BCL11B*, narrowed by RP11-74H1 (centromeric) and RP11-431B1 (telomeric) probes. (D) Breakpoints on 16q24 spanned a 330.1 kb region centromeric to *FOXF1*, between clones RP11-592D21 (centromeric) and RP11-1055L9 (telomeric). Both panels C and D are not to scale, and the breakpoints shown correspond to those inspected and confirmed using IGV. (E-F) WGS coverage tracks of samples from SJALL080601 and SJALL070948 (E) and SJALL080603 (F) showing the duplication of the der(16) with (SJALL080601 and SJALL070948) or without (SJALL080603) the concomitant loss of the normal chromosome 16. Dark blue tracks represent chromosome 14 and light blue tracks represent chromosome 16. Subclonal events in case SJALL080601 were found by FISH/SNP array. Schematic representations of the patients' rearrangements are shown on the right. (G-H) H3K27ac HiChIP interaction maps generated using patient-specific genomes containing the rearranged chromosome sequences in the primary sample SJALL079622 (G) and the PDX from case SJTALL005179 (H). In both panels, the derivative chromosome 14 is shown. Insets show schematic representations of the der(14). Significant H3K27ac-anchored interactions (false discovery rate < 0.01) are shown as arcs. All coordinates are hg38. E, enhancer; bp, breakpoint.

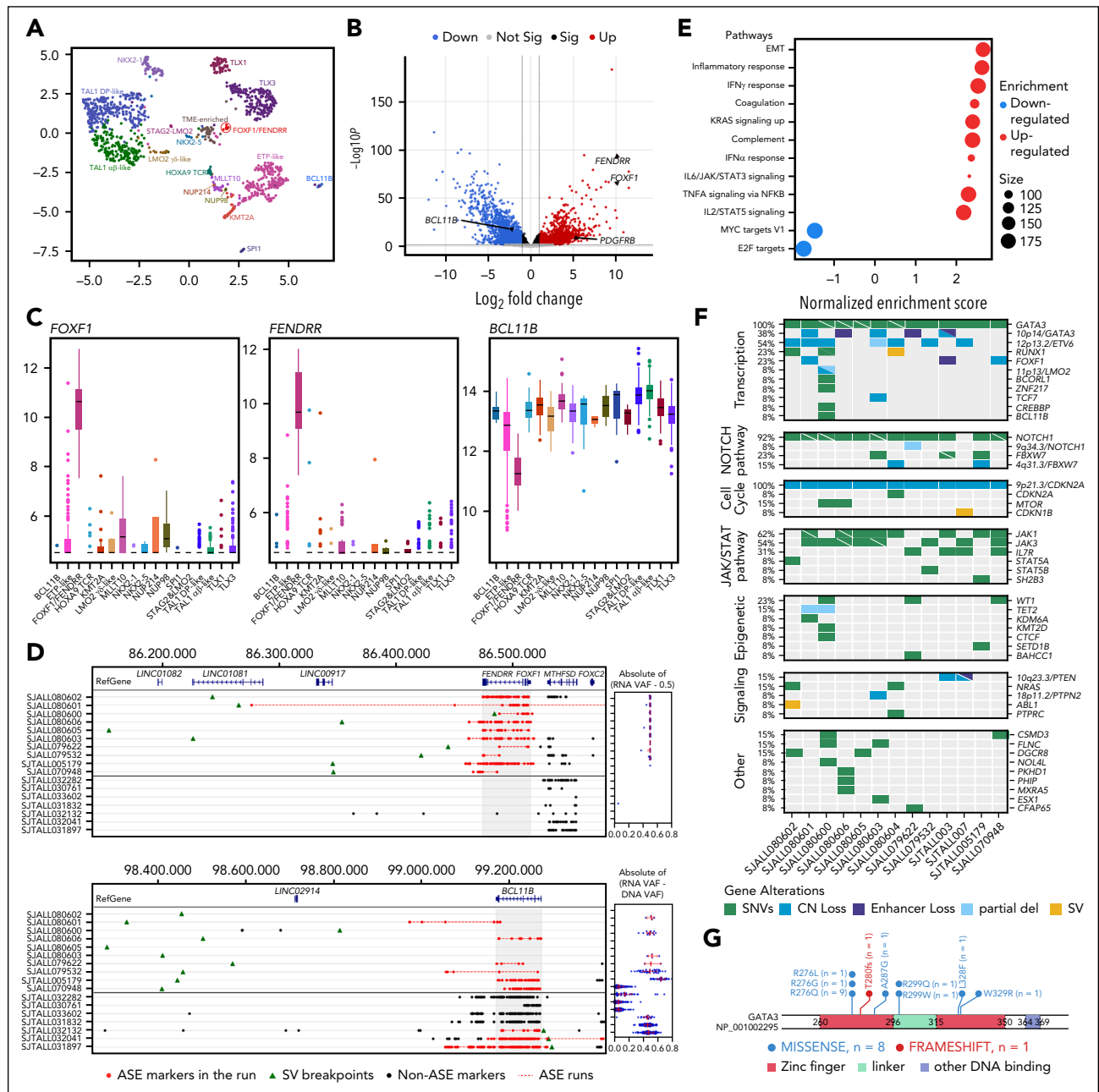


Figure 2. *FOXF1/FENDRR* ALL is a distinct transcriptional subtype with defined genetic co-lesions. (A) uniform manifold approximation and projection (UMAP) plot showing gene-expression analysis of 1370 T-ALL samples, including 10 *FOXF1/FENDRR* samples (circled). The same features as in Pölönen et al⁴ were used for the UMAP projection. (B) Volcano plot showing the gene-expression signature of *FOXF1/FENDRR* samples. Black dots indicate genes with a significant false discovery rate (≤ 0.05). Red and blue represent upregulated and downregulated differentially expressed genes, respectively (false discovery rate ≤ 0.05 , $\log_2$ fold change ≥ 1). (C) Boxplots showing the expression levels (variance-stabilizing transformation, vst) of *FOXF1*, *FENDRR* and *BCL11B* across leukemic subgroups. (D) Allele-specific expression (ASE) analysis of *FOXF1/FENDRR* (top panel) and *BCL11B* (bottom panel). Each row represents a leukemia sample with matched WGS and RNA-seq data, allowing allele frequency analysis at heterozygous SNPs. Left panels: red dots mark SNPs with significant allelic imbalance in RNA-seq; dashed red lines indicate continuous runs of such SNPs; black dots denote SNPs without significant imbalance. Right panels: for *FOXF1/FENDRR*, located within a region of 16q loss (from *FENDRR* to *FOX1*), the plot shows the absolute deviation of RNA variant allele frequency (VAF) from 0.5. For *BCL11B*, located in a region of 14q gain, the plot displays the absolute difference between RNA-seq and WGS VAF, highlighting expression imbalances due to copy number alterations. Four samples from the *TLX1* subgroup (*TRA::TLX1*: SJTALL032282, SJTALL030761, and *TRB::TLX1*: SJTALL033602, SJTALL031832) and 3 samples from the *BCL11B*-a subgroup (*BCL11B::ARID1B*: SJTALL032132, SJTALL032041, SJTALL031897) were used as controls. Among the *FOXF1/FENDRR* samples, ASE analysis for *BCL11B* was informative only for those samples with detectable gene-expression (SJALL080601, SJALL080606, SJALL079622, SJALL079532, SJALL005179, SJALL070948). (E) GSEA showing top 10 up- and downregulated pathway in the t(14;16) samples vs other leukemic genetic subgroups (EMT, epithelial-mesenchymal transition). (F) Oncoprint showing the additional genomic alterations characterizing samples with t(14;16). Only genes that are recurrent (present in more than one sample) or identified as significant in Pölönen et al⁴ are included. *STAT5A* was added due to the frequent involvement of the JAK/STAT signaling pathway. Variants are reported from diagnosis and relapse stages for samples SJALL079532 and SJALL080602, respectively. (G) Schematic representation of the GATA3 DNA-binding domain (DBD), with mutation sites identified in the patient cohort indicated. (H) UMAP scatterplot showing the distribution of GATA3 mutations across the 1370 T-ALL samples investigated by RNA-seq. (I) Violin plots showing the expression levels (vst) of genes involved in T-cell fate regulation in the *FOXF1/FENDRR* subgroup compared to other T-ALL subtypes. Statistical significance was assessed using the Wilcoxon rank-sum test. Asterisks indicate levels of significance: $P \leq .001$ (***), $P \leq .0001$ (****).

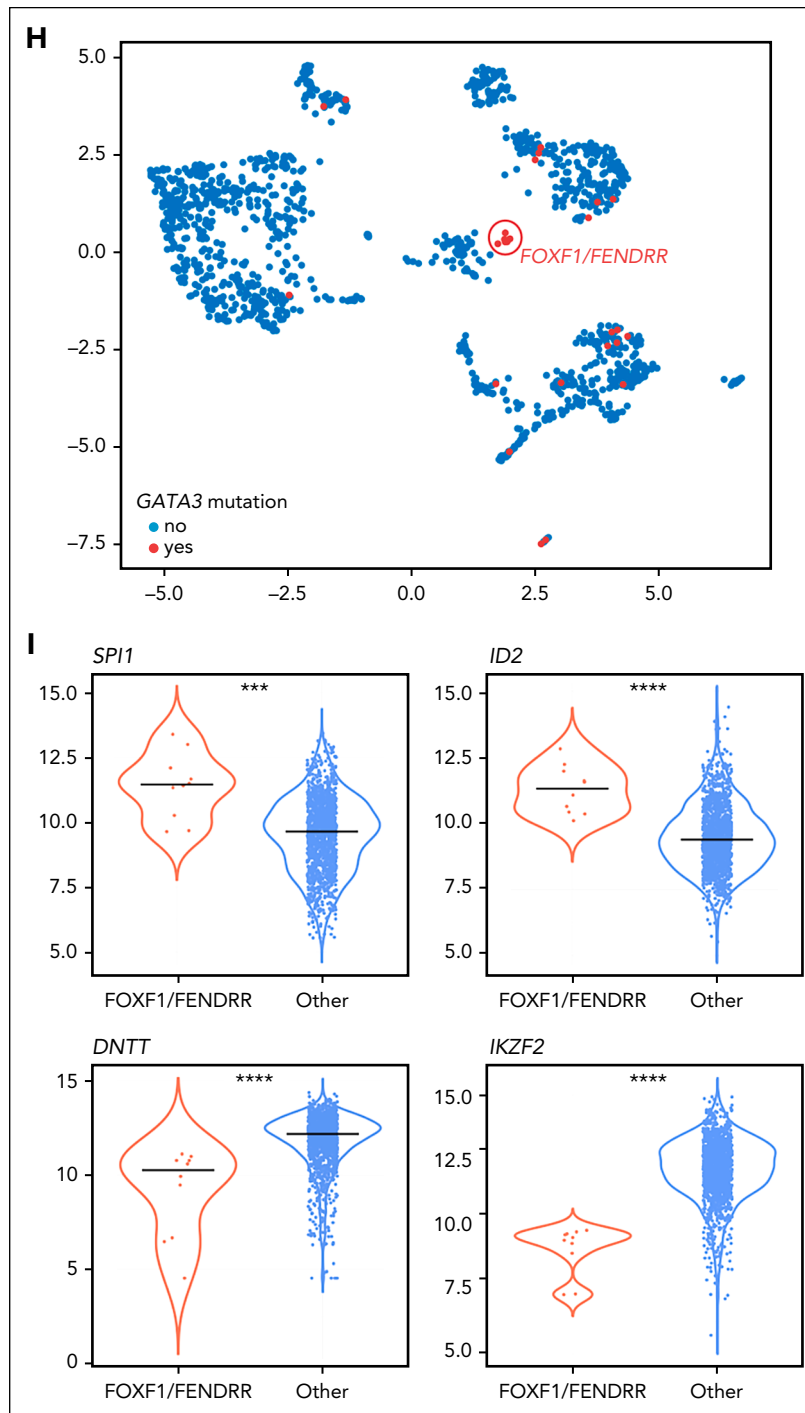


Figure 2 (continued)

FOXF1 dependency and genomic occupancy sustain the gene-expression program of FOXF1/FENDRR leukemia

To investigate whether this novel subtype depends on *FOXF1* overexpression, we established a cell line from the SJALL079622-derived PDX. This model faithfully recapitulated genomic features of this subtype and retains the B/T lymphoid immunophenotype of the primary sample and the PDX (Figure 3J). Using a CRISPR/Cas9 approach to inactivate

FOXF1 we observed a marked growth disadvantage, supporting the oncogenic role of its overexpression in this context (Figure 3K). We confirmed *FOXF1* overexpression in cells from 2 PDX models (SJALL079622 and SJTALL005179; Figure 4A) and we explored its genome-wide occupancy. Analysis of ChIP-seq data revealed 24 111 significant peaks shared between the 2 samples (supplemental Table 23), with *FOXF1* most commonly binding at promoter regions (49.5% of peaks; Figure 4B). Previous experiments in gastrointestinal stromal

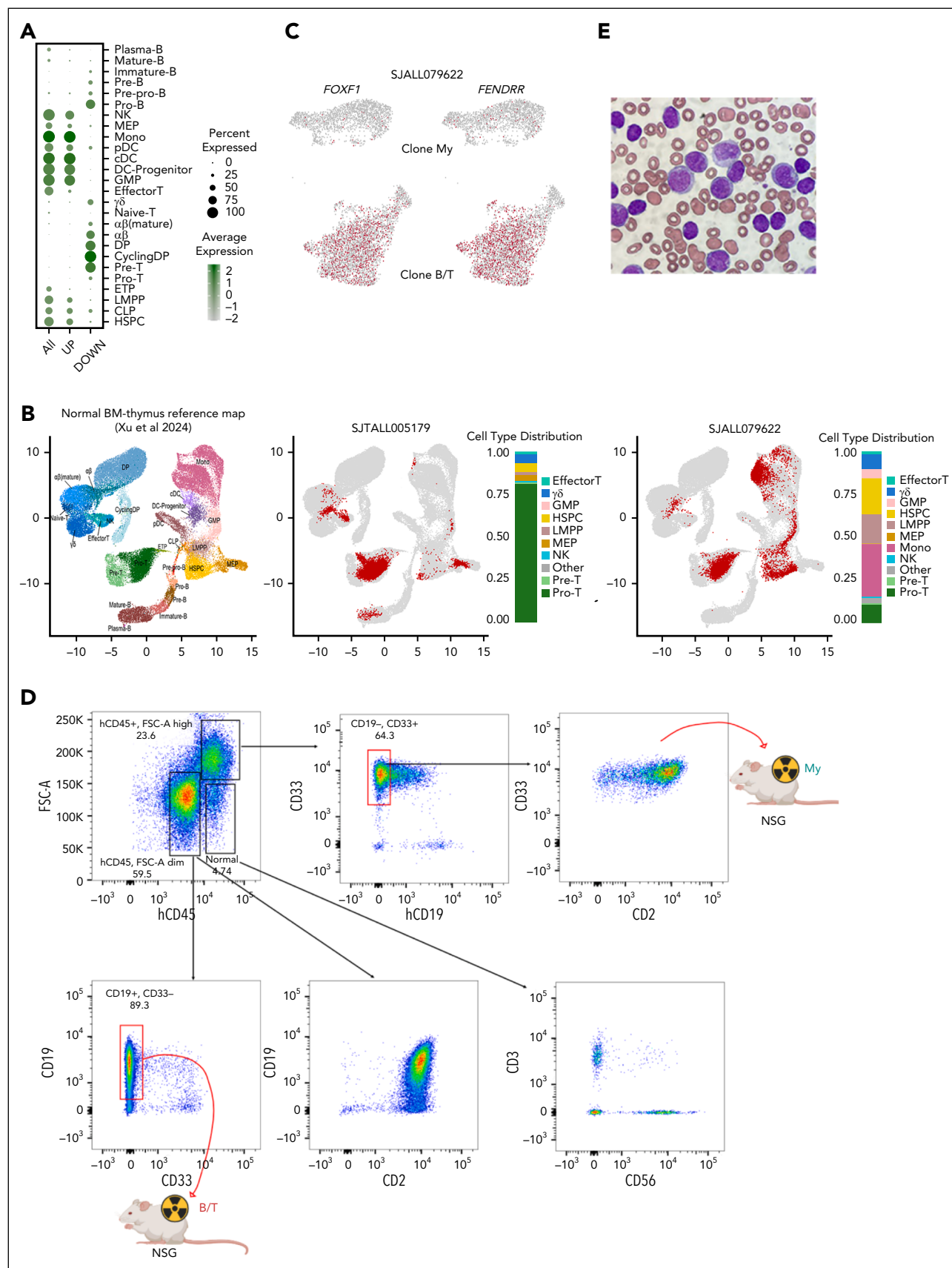


Figure 3. Transcriptional and genetic intrasample and intersample heterogeneity in FOXF1/FENDRR ALL. (A) Gene set scoring of the FOXF1/FENDRR-positive subtype signature in normal BM and thymic scRNA-seq data. The full set of differentially expressed genes (DEGs) was scored. The dot plot depicts the percentage of

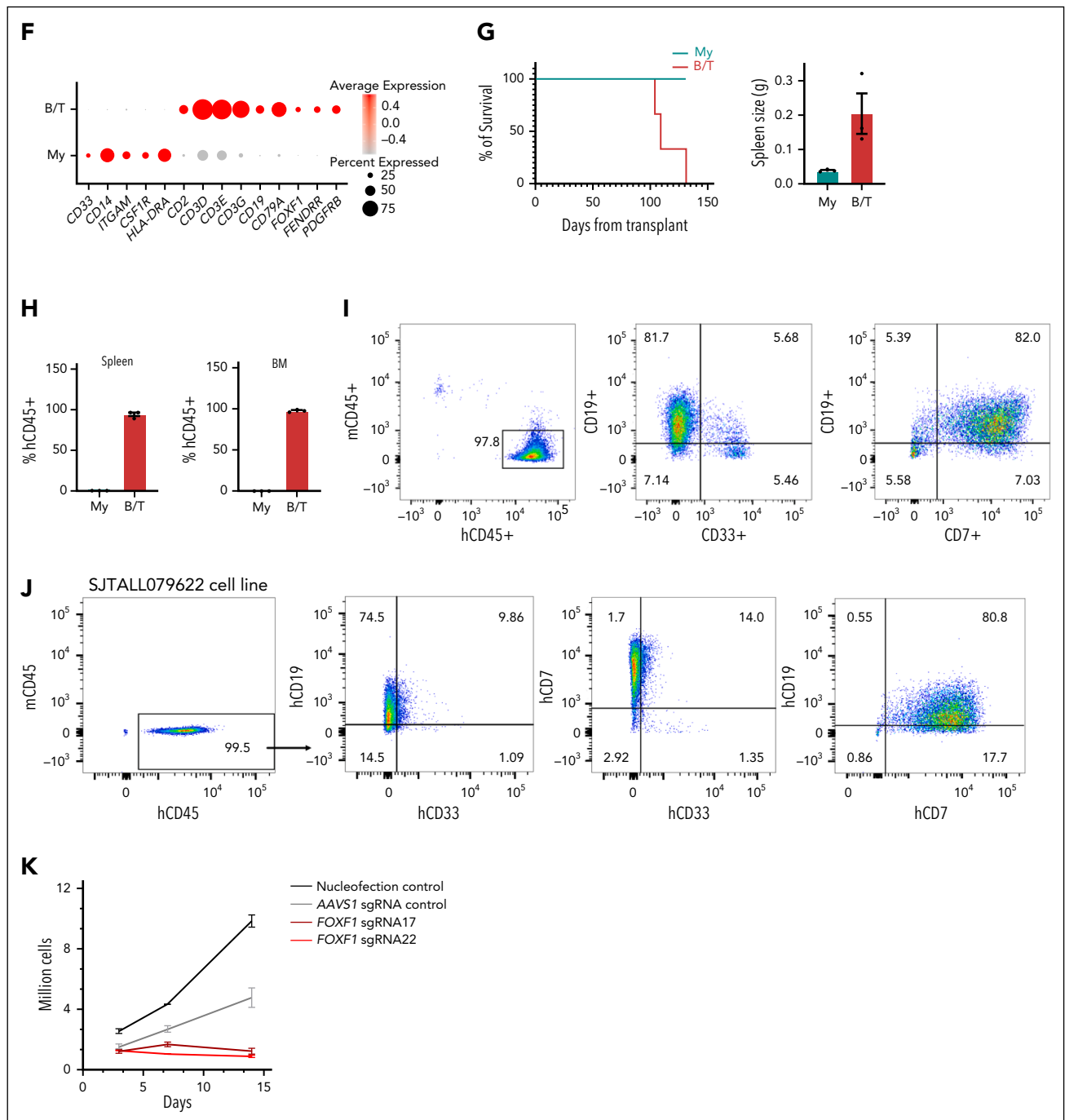


Figure 3 (continued) signature detection (dot size) and the mean gene set score by normal cell type (color). (B) scRNA-seq data for primary samples SJTALL005179 and SJTALL079622 were projected onto the map of normal BM and thymus derived hematopoietic cells¹⁹ to visualize enrichment for different developmental states. The stacked bar plot at the right of each panel quantifies the proportion of leukemic cells mapping to each developmental state. (C) UMAP plot showing the expression of *FOXF1* and *FENDRR* in the leukemic blasts from sample SJTALL079622, profiled by scRNA-seq. (D) Flow cytometry results from the primary sample SJTALL079622 showing co-existing CD33⁺ CD19⁺ (indicated as “My,” myeloid clone) and CD33⁺ CD19⁺ (indicated as “B/T,” lymphoid clone) populations in the BM, which were isolated by cell sorting and independently injected into recipient immunocompromised mice. Part of the figure panel has been created with [biorender.com](https://biorender.com/zym1mqn). Di Giacomo, D. (2026) <https://biorender.com/zym1mqn>. (E) PB smear from sample SJTALL079622 showing the co-occurrence of myeloid and lymphoid blast populations. (F) Dot plots showing enrichment of myeloid-associated genes in one clone (My) and lymphoid-associated genes in a second clone (B/T), highlighting the transcriptional heterogeneity and multiple lineage involvement within the leukemic blasts. Key genes from the *FOXF1*/*FENDRR* leukemic subtype signature are enriched in the lymphoid clone. (G) Left: Kaplan-Meier survival curves of recipient mice transplanted with sorted populations from sample SJTALL079622; right: leukemic burden assessed by spleen weight. (H) Cell engraftment measured as the percentage of hCD45⁺ cells in the spleen and BM. (I) Flow cytometry showing engraftment of the lymphoid-sorted population in the BM of a representative mouse. (J) Flow cytometry showing the expression of B and T lymphoid markers in the SJTALL079622 PDX-derived cell line, SJT-622-5. (K) Growth curves of SJT-622-5 cells following RNP-mediated CRISPR/Cas9 inactivation of *FOXF1*. Cell proliferation was monitored over 15 days after electroporation with 2 independent sgRNAs targeting *FOXF1* (sgFOXF1 nos 17 and 22). A nontargeting sgRNA against the *AAVS1* locus and a no-nucleofection control were included to account for electroporation-related toxicity and baseline growth, respectively. Samples were tested in duplicate and data are reported as mean $\pm$ standard deviation (SD). Data are shown for one of two independent experiments; these had concordant results showing dependence of the cell line on *FOXF1* expression.

A

B

C

D

E

F

G

tumor cell lines dependent on FOXF1 showed a similar binding pattern,²⁰ although with fewer promoter sites and more intronic sites (Figure 4C). This suggests that FOXF1 engages different regulatory programs depending on cellular context.

Integrative analysis of ChIP-seq data with H3K27Ac HiChIP and ATAC-seq revealed that FOXF1 binds to regions of accessible

chromatin and is associated with markers of active chromatin state (Figure 4D). FOXF1 binding sites could be subdivided into those accompanied by H3K27 acetylation and/or chromatin accessibility as evidenced by ATAC-seq. We then performed single-sample GSEA using the top 500 peaks from FOXF1 ChIP-seq data to assess the activity of FOXF1-regulated genes across different subtypes. The *FOXF1/FENDRR* subtype exhibited

higher single-sample GSEA enrichment scores compared to all other subtypes, indicating increased activity of FOXF1 target regions in this group (Figure 4E).

Among FOXF1-bound regions, the *FOXF1/FENDRR* locus displayed prominent occupancy (Figure 4F), supporting the existence of a feed-forward regulatory loop between FOXF1 and its lncRNA partner *FENDRR*.²¹ In addition, we observed strong enrichment of FOXF1 binding at the *PDGFRB* locus (Figure 4G), which correlates with its elevated expression in the *FOXF1/FENDRR* leukemia subgroup (supplemental Table 10; Figure 2B). Together, these findings suggest that in this specific leukemic context, FOXF1 contributes to gene regulation by binding to active cisregulatory elements and promoting expression of its target genes.

Clinical features of *FOXF1/FENDRR* leukemia

Patients with *FOXF1/FENDRR* leukemia were children ($n = 3$) or young adults ($n = 10$) (median, age 19; range, 7-37) with male predominance (male:female, 10:3). Extramedullary involvement of the mediastinum or pleura, and/or central nervous system infiltration was observed in 10 of 13 patients. Immunophenotype was available for 11 of 13 patients (Table 1). Based on the immunophenotypic assessment, patients were initially diagnosed as ETP-ALL ($n = 7$), T/myeloid MPAL ($n = 1$), B/T MPAL ($n = 3$); 2 patients were classified as having ETP-like leukemia (Table 1). Shared immunophenotypic features consisted in positivity for CD38, cCD3, CD7 and CD13, and negativity for CD34, CD1a, CD4; TdT was negative in 9 of 11 and CD8 in 8 of 11. A distinctive hallmark was the expression of at least 1 B-lymphoid marker (CD10, cyCD79 and/or CD19) found in 10 of 11 samples. Some immunophenotypic features, such as CD13 upregulation and downregulation of CD34, CD1a, and DNTT (TdT), were also corroborated by RNA-seq (supplemental Table 10).

A common feature was poor response to front line and subsequent therapy (10 patients), and relapse following hematopoietic stem cell transplantation ($n = 3$). One case (SJALL080604) was refractory to 4 lines of therapy, and transient hematological and cytogenetic remission was achieved with the tyrosine kinase inhibitor (TKI) dasatinib, administered as a single agent due to the presence of an *FXR1::ABL1* fusion. In a second case (SJALL080602), ponatinib was administered based on in vitro drug response profiling and induced partial hematological and cytogenetic remission, allowing the patient to be enrolled in a hematopoietic stem cell transplant program; he achieved complete remission lasting 4 months. Twelve patients died from disease progression; only 1 patient (SJALL080605) is still alive 48 months after diagnosis and 42 months from hematopoietic stem cell transplant.

Sensitivity to BCL family and JAK/STAT pathway inhibitors

Given the poor outcome of this subtype, we investigated therapeutic vulnerabilities by performing ex vivo drug screening using a library of 176 compounds (supplemental Figure 4A) on leukemic cells isolated from PDXs. Extensive cell death in untreated control cells prompted exclusion of SJTALL005179-PDX from the analysis. A total of 65 and 56 compounds met sensitivity criteria (Drug Sensitivity Score, ≥ 10 ; and area under

the curve, <300) in SJALL080600 and SJALL079622, respectively (supplemental Figure 4B), with 46 drugs shared between the two (supplemental Table 25). They included kinase inhibitors (18/46, 39.1%), conventional chemotherapeutics (13/46, 28.3%), epigenetic modifiers (6/46, 13%), apoptotic modulators (5/46, 10.8%), and NOTCH1 and proteasome inhibitors (2/46, 4.4%) (supplemental Table 26). Among kinase inhibitors, enrichment of compounds targeting the PIM kinases family members and the PI3K/mechanistic target of rapamycin (mTOR) pathway were mechanistically relevant. PIM kinases are transcriptional targets of phosphorylated STAT factors, and mTOR is an effector of the JAK/STAT pathway in sustaining T-ALL metabolism and growth.²² Together these pathways contribute to leukemic cell survival by promoting antiapoptotic signaling.²²⁻²⁵ Consistent with this, we observed elevated expression of both PIM kinases and antiapoptotic BCL-2 BCL2 apoptosis regulator (BCL-2) family members in *FOXF1/FENDRR* leukemia compared to other T-ALLs (supplemental Figure 4C).

SJTALL005179 and SJALL079622 samples were assessed for cellular dependency on antiapoptotic proteins and sensitivity to JAK/STAT and PI3K/mTOR pathway inhibition. Splenic cells from PDXs were treated with BH3-only peptides specifically targeting BCL-2, BCL-xL, or MCL-1 followed by assessment of mitochondrial depolarization. Both models depended on BCL-xL and MCL-1, as evidenced by HRK- and MS2-induced depolarization, respectively, whereas PDX SJALL079622 exhibited additional BCL-2 dependence, showing greater depolarization in response to the BAD peptide than to HRK (Figure 5A; supplemental Figure 4D).

To examine the utility of inhibiting the JAK/STAT and mTOR pathways, we assessed the basal phosphorylation status of their key signaling components. We observed constitutive phosphorylation of STAT5 in both PDX models, whereas S6 phosphorylation was restricted to SJALL079622 (supplemental Figure 4E); this reflects the presence of an activating *IL7R* mutation, which partially signals via PI3K/mTOR.²² As further confirmation, ex vivo treatment with the JAK1/2 inhibitor ruxolitinib resulted in reduced basal STAT5 phosphorylation in both models (Figure 5B), whereas the mTOR inhibitors rapamycin and temsirolimus similarly decreased pS6 levels in SJALL079622 (Figure 5C). Validating the drug sensitivity findings on PDX SJALL079622, we observed sensitivity to the BCL-2/BCL-xL inhibitors navitoclax (ABT-263) and venetoclax (ABT-199),^{26,27} with navitoclax inducing a more complete cytotoxic response at higher concentrations as a single agent (Figure 5D). Notably, both inhibitors exhibited a synergism with either ruxolitinib or the pan-PIM inhibitor PIM447 (Figure 5E; supplemental Figure 4F). In contrast, although mTOR inhibitors were effective as single agents (Figure 5F), no significant synergism was observed when combined with JAK/STAT inhibition (Figure 5G; supplemental Figure 4G). Likewise, standard-of-care agents showed no synergism, either combined with each other or with BCL-2 inhibitors (supplemental Figure 4H).

These results suggest a therapeutic approach pairing antiapoptotic inhibitors with JAK/STAT inhibitors to exploit synergy and improve efficacy, and align with previous preclinical studies demonstrating the efficacy of JAK/STAT inhibition in ETP-ALL.^{28,29}

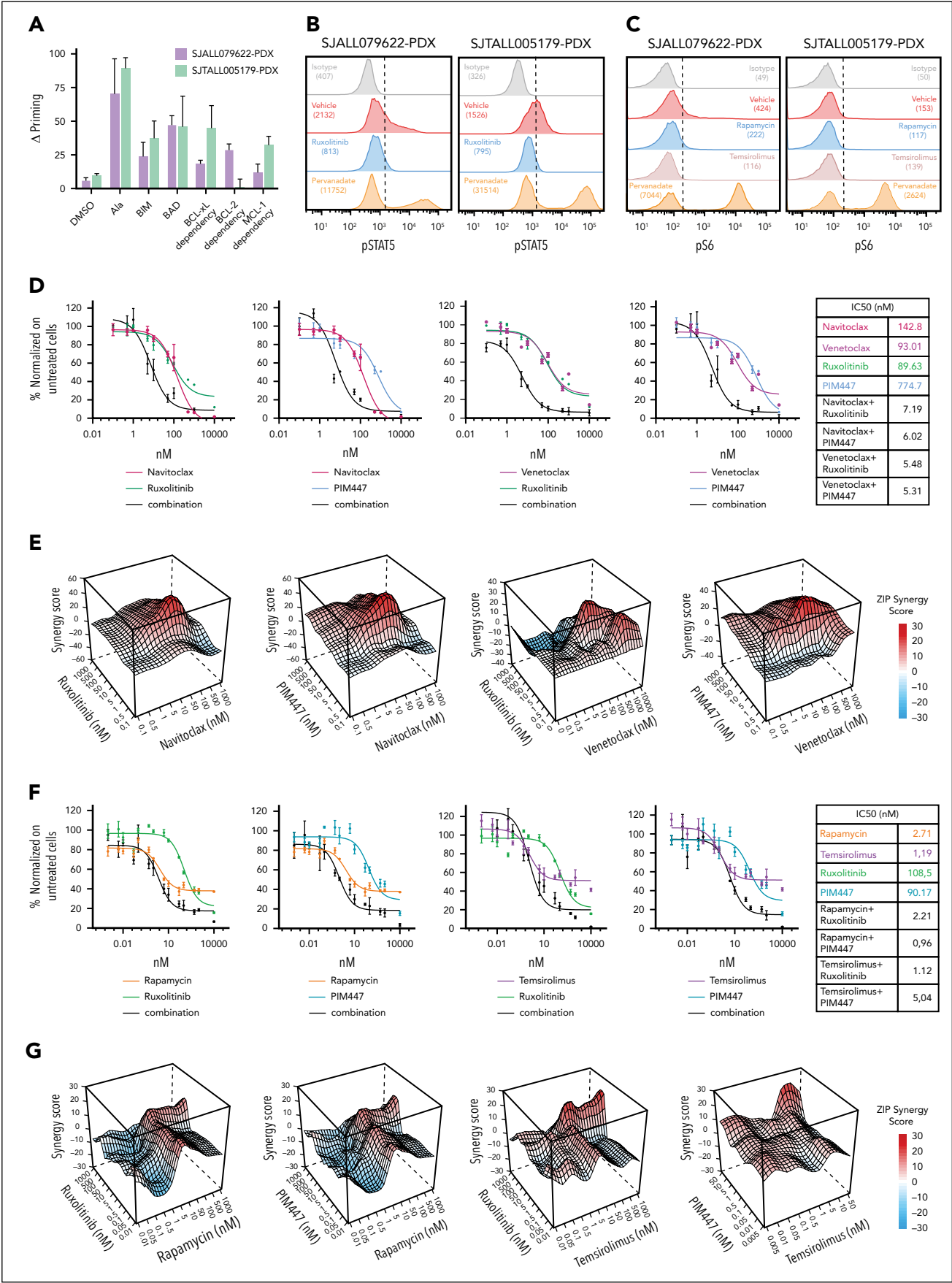


Figure 5.

Discussion

In this study, we describe a t(14;16)(q32;q24) translocation that deregulates *FOXF1/FENDRR* expression through juxtaposition to *BCL11B* regulatory elements, and that defines a subgroup of high-risk lineage ambiguous leukemia of variable immunophenotype with a distinct gene-expression profile.

FOXF1 (Forkhead box protein F1) is a forkhead box (FOX) transcription factor expressed the lung, placenta, and prostate.³⁰ In the lung, *FOXF1* expression is regulated by a ~272 kb upstream enhancer.^{31,32} *FOXF1* and lncRNA *FENDRR*, located ~1.7 kb upstream, are examples of divergent transcription: they are transcribed in opposite directions, and they share a genomic region containing promoters. In the human lung, *FENDRR* expression is regulated in *cis* by the *FOXF1* enhancer and in *trans* through direct binding of *FOXF1* to its promoter.^{33,34} *FENDRR* functions as an epigenetic modifier by recruiting histone methyltransferase complexes to gene promoters, and acting as a molecular sponge for miRNAs, including *miR-424* which is a negative regulator of *FOXF1*, thereby indirectly promoting *FOXF1* expression. This regulatory loop demonstrates how *FOXF1* and *FENDRR* mutually enhance expression through transcriptional and epigenetic mechanisms.

Neither *FOXF1* nor *FENDRR* are physiologically expressed in the hematopoietic system. However, the t(14;16)(q32;q24) replaces the lung *FOXF1-FENDRR* distal enhancer with *BCL11B* regulatory elements, resulting in their ectopic expression. This represents a broader mechanism in T-ALL, where lineage-specific enhancers are recurrently hijacked to activate developmentally silenced genes involved in embryonic or organ-specific programs.⁴ While the functional relevance of *FENDRR* remains to be elucidated, the central regulatory role of *FOXF1* is supported by the coordinated upregulation of its downstream targets within the subgroup-specific transcriptional program and further validated by the reduced proliferative capacity of *FOXF1*-silenced cells. The activation of both genes in t(14;16)-negative immature T-ALL suggests that alternative regulatory mechanisms, such as epigenetic or upstream transcriptional programs, may underline their expression and warrant further investigation.

GATA3 loss of function alterations are a universal feature of *FOXF1/FENDRR* leukemia. *GATA3* has a central role in thymic T-cell maturation, including progression of early T-cell precursor through to the double negative 2 stage,³⁵ regulating T-cell commitment through priming *BCL11B* expression,^{36,37} silencing of *SPI1/PU.1*, repression of NOTCH targets such as *DTX1*, *CD3E*, *IL7R*, and *NOTCH1/3*, promoting the upregulation of genes such as *MYC* and *HES1*¹⁸ and extinguishing alternative fates such as B-cell and NK-cell differentiation^{18,38}

Our results suggest that inactivation of *GATA3* has a role in the immunophenotypic and transcriptomic perturbations observed in *FOXF1/FENDRR* leukemia, by impairment of *BCL11B* activation and *SPI1* silencing, resulting in the arrest of T-cell precursors at a precommitment stage. This developmental block, further evidenced by the expression of B-cell markers and the persistence of immature and non-T-lineage transcriptional programs, may be reinforced by alterations in the cell cycle regulator gene *CDKN2A*, which modulate the balance between apoptosis and proliferation within the continuum of thymic immature-cell differentiation.³⁷

Indeed, our prior studies of early T-cell precursor ALL³ and childhood T-ALL⁹ identified *GATA3* mutations as a common feature of ETP-ALL, but lacked the breadth of leukemia subtypes, and integrated multiomic approaches required to define *FOXF1/FENDRR*-activating rearrangements and *GATA3* alterations as the universal features of a distinct subtype of leukemia.³⁹ This highlights the critical need for precise diagnostic approaches to accurately detect such rearrangements. In *FOXF1/FENDRR* ALL, the 14q32 breakpoints localized within the same cluster as other 14q32/*BCL11B* translocations in T-ALL known to activate oncogenes.^{10,11} This cluster is distinct from the *BCL11B*-a acute leukemia breakpoint region and can be reliably distinguished by FISH,⁷ which thus offers a potential diagnostic tool.

Combination therapies based on antiapoptotic inhibitors targeting the BCL-2 family proteins and JAK/STAT inhibitors may represent a promising strategy in *FOXF1/FENDRR* leukemia. In T-ALL, the maturation stage of leukemic blasts appears to influence their dependence on BCL-2 family proteins for survival. Preclinical studies using cell lines and xenograft models⁴⁰ have shown that typical T-ALL primarily depends on BCL-xL and expresses higher levels of *BCL2L1*, whereas ETP-ALL relies on BCL-2 and shows higher *BCL2* expression.³⁹ Although the BCL-2/BCL-xL inhibitor navitoclax is limited by on-target thrombocytopenia due to BCL-xL inhibition⁴¹ acquired resistance represents the main limitation for venetoclax.^{42,43} Therefore, novel approaches based on BCL-xL degraders,⁴⁴ or combination strategies including multiple BH3-mimetic drugs or rationally selected agents (ie, JAK/STAT inhibitors)⁴⁵ may be an attractive approach to enhance treatment efficacy and overcome resistance.⁴³ Notably, a recent clinical trial in patients with relapsed/refractory myelofibrosis demonstrated that the combination of navitoclax and ruxolitinib provided durable clinical benefit with a generally manageable safety profile.⁴⁴ An intriguing observation in this subgroup is the high expression of *PDGFRB*, with *FOXF1* binding detected at its locus. Notably, 2 samples showed sensitivity to TKIs, suggesting these agents may represent an additional therapeutically exploitable vulnerability.

Figure 5. BCL-2 family and JAK/STAT inhibition in *FOXF1/FENDRR* leukemia. (A) Mitochondrial depolarization levels induced by individual BH3 mimetics across 2 PDX models. The bar plot shows delta priming values (% cytochrome c release [treatment] – % cytochrome c release [DMSO]), representing the percentage of PDX blasts in the cytochrome c-negative gate following peptide exposure. Data are reported as mean ± standard error of the mean from 2 independent experiments for each mouse. Ala, alamethicin was used as technical positive control. The BIM peptide reflects the overall degree of apoptotic priming, while BAD-induced depolarization indicates dependence on antiapoptotic BCL-2 family members. Dependence on BCL-xL and MCL-1 was determined using HRK and MS1 peptides, respectively. BCL-2 dependence was calculated by subtracting the effect of HRK (specific for BCL-xL) from that of BAD. (B-C) Phosphoflow cytometric analysis of phosphorylated STAT5 (B) and pS6 (C) in the 2 indicated PDX models. Numbers in brackets indicate the mean values of Alexa-647 intensity signal for each condition. (D) Dose-response curves for navitoclax/ABT-263 and venetoclax/ABT-199 in combination with ruxolitinib, and PIM447 are shown for PDX from patient SJALL079622. Black curves indicate the combination of the 2 drugs in each graph. One of 2 independent experiments is reported. (E) 3D heat maps depict the synergy of each drug combination shown in the corresponding curve in panel D, with synergy scores calculated using the zero interaction potency method. (F) Dose-response curves for the mTOR inhibitors rapamycin and temsirolimus in combination with ruxolitinib or PIM447 in the same PDX model. Black curves indicate drug combinations. (G) Corresponding 3D synergy heat maps for the combinations shown in panel F, with zero interaction potency synergy scores. DMSO, dimethyl sulfoxide.

In conclusion, our identification of the *FOXF1/FENDRR* subgroup as a genetic entity of acute leukemias underscores the importance of genomic classification to enhance our understanding of leukemia biology and improve patient stratification. The prognostic and predictive significance of *FOXF1/FENDRR* prompts the introduction of specific diagnostic tests into the diagnostic evaluation of immature acute leukemia workflow, enabling early identification of this high-risk subgroup, and guiding the development of alternative treatment regimens that exploit its unique molecular vulnerabilities.

Acknowledgments

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Authorship

Contribution: D.D.G. led the study, performed experiments, analyzed genomic and expression data, and wrote the manuscript; P.P. analyzed whole-genome sequencing and single-cell data; V.B. performed custom next-generation sequencing- and gene-expression experiments; S.K. contributed to sample collection and data analysis; S.A., C.N., and A.G.L.F. contributed to gene-expression studies; V.P. performed and analyzed fluorescence in situ hybridization and SNP-array experiments; L.P. and G.R. performed drug-screening profiling; Y.C. performed ChIP-seq experiments; Q.G. and W.R. analyzed ChIP-seq data; C. Matteucci performed custom next-generation sequencing experiments; L.E.M. contributed to HiChIP and BH3 profiling data analysis; Y.W. contributed to cell line generation and *FOXF1* dependency validation; C.S.P. and H.W. contributed to the *FOXF1* dependency validation studies; W.C.W. performed high-throughput in vitro drug combination screening; F.V. contributed to the drug-screening validation; A.Q. analyzed MissionBio data; E.B.W. worked with patient-derived xenograft (PDX) models; P.B. contributed to single-cell experiments; M.C., M.L., A. Manabe, S.H., J.T., D.H., S.M., and D.T.T.

collected samples and patient data; A. Macchiarulo contributed to the interpretation and contextualization of mutational data; J.X. contributed to single-cell RNA sequence analysis; I.I. performed single-cell experiments, established PDX models, and contributed to manuscript preparation; R.L.S. contributed to study design, collected samples and patient data, and cowrote the manuscript; C. Mecucci and C.G.M. conceived, designed, and supervised the study, and cowrote the manuscript; and all authors approved the final version of the manuscript.

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Footnotes

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Genomic data generated in this study have been deposited in the European Genome Phenome Archive (accession number EGAS50000001255).

The online version of this article contains a data supplement.

There is a [Blood Commentary](#) on this article in this issue.

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